



SCAN_GO

Smart Supermarket Checkout System

Project Members

Mahmoud Osama Elsaid

Mazen Barakat

Abdallah Mostafa

Ahmed Wagih

Mahmoud Atef

Sohaib Hamada

Supervised by

Dr. Mohamed Kamal

Assistant

Eng. Doria Gamal

2025/2026

Acknowledgement

We would like to express our sincere gratitude to everyone who supported us throughout the development of this graduation project, **SCAN_GO – Smart Supermarket Checkout System**.

First and foremost, we extend our deepest appreciation to **Dr. Mohamed Kamal** for his invaluable guidance, continuous support, and insightful feedback, which played a crucial role in shaping this project.

We would also like to thank **Eng. Doria Gamal** for her assistance, encouragement, and technical support throughout the different phases of the project.

We are also grateful for the collaborative efforts of our team members, whose dedication and teamwork were essential in successfully completing this project. Special thanks to all individuals who provided constructive feedback and valuable suggestions that helped improve the quality of our work.

We also acknowledge the technological resources and tools that facilitated the development and implementation of this system.

Our appreciation also goes to the faculty members of the Computer Science Department at Misr Higher Institute for Commerce & Computers for providing us with the knowledge and resources necessary to complete this work.

Finally, we would like to thank our families and friends for their constant support, patience, and motivation during this journey.

Contents

Acknowledgement	2
ABSTRACT	5
Chapter One	6

1.1 Background and Context	7
1.2 Problem Statement	8
1.2.1 Long Waiting Times at Checkout	10
1.2.2 Inefficiency of Manual Checkout Processes	10
1.2.3 Complications Associated with Cash Transactions	11
1.2.4 High Operational Costs	11
1.2.5 Lack of Real-Time Inventory and Sales Monitoring	11
1.2.6 Inventory Shrinkage and Security Issues	12
1.2.7 Poor Customer Experience	12
1.2.8 Limitations of Existing Smart Retail Solutions	12
1.2.9 Summary of the Problem	13
1.3 Aim of the Project	13
1.4 Project Objectives	15
1.4.1 Development of a Mobile-Based Self-Scanning System	15
1.4.2 Implementation of a Digital Payment System	16
1.4.3 Design and Development of a Backend System	16
1.4.4 Creation of an Administrative Dashboard	16
1.4.5 Implementation of QR Code-Based Verification	17
1.4.6 Integration of System Components	17
1.4.7 Ensuring System Scalability	18
1.4.8 Secure and Efficient Inventory Management	18
1.5 Proposed System Overview	18

1.6 System Architecture Overview	20
1.6.1 Mobile Application Layer	21
1.6.2 Backend System Layer	21
1.6.3 Database Layer	22
1.6.4 Admin Dashboard Layer	23
1.6.5 Communication Layer (API Integration)	23
1.6.6 Security Considerations in Architecture	23
1.6.7 Scalability and Performance	24
1.6.8 Summary of System Architecture	24

ABSTRACT

This project presents SCAN_GO, a smart basket-based checkout system designed to eliminate traditional cashier-based processes and improve the overall shopping experience. The system is inspired by cashier-less retail concepts such as Amazon Go but is specifically designed as a cost-effective and scalable solution suitable for developing environments. SCAN_GO operates by integrating a tablet, a microcontroller, a barcode scanner, and a weight sensor directly onto the shopping basket, allowing customers to scan items while automatically validating transactions through real-time weight verification. Unlike traditional systems that rely on manual checkout procedures, this approach effectively eliminates waiting times and enhances operational efficiency. The system consists of three main components: an Android application for customer interaction, a centralized backend API for handling business logic and data processing, and an administrative dashboard for real-time monitoring of sales and inventory. The proposed solution addresses several key challenges in conventional retail environments, including long checkout queues, inefficiencies in manual processing, and retail loss. Additionally, SCAN_GO incorporates a QR-code-based exit verification gate to ensure secure, accurate transactions, improved customer satisfaction, and enhanced inventory control. Overall, SCAN_GO provides a practical, highly scalable approach to modernizing retail operations.

CHAPTER 1 (Introduction)

1.1 Background and Context

In recent decades, the retail industry has experienced a rapid transformation driven by technological advancements and evolving consumer expectations. Traditional shopping experiences, which once relied heavily on manual interactions between customers and cashiers, are gradually being replaced by automated and digital solutions. This transformation is not only a response to technological innovation but also a necessity to remain competitive in an increasingly fast-paced and customer-centric market.

Supermarkets, in particular, represent a critical component of daily life, serving a wide range of customers with diverse needs. However, despite technological progress in other sectors, many supermarkets still depend on outdated checkout systems. These systems are characterized by long queues, manual barcode scanning, and reliance on physical cash transactions. Such inefficiencies result in reduced customer satisfaction and increased operational costs.

With the rise of smartphones, cloud computing, and digital payment systems, new opportunities have emerged to revolutionize the retail experience. Modern consumers demand speed, convenience, and flexibility, expecting seamless interactions without unnecessary delays. This shift in expectations has encouraged the development of innovative retail concepts such as cashierless stores.

One of the most notable examples is Amazon Go, which introduced a fully automated shopping experience using advanced technologies such as computer vision, machine learning, and sensor fusion. While this model represents a significant leap forward, it is associated with high implementation costs due to its reliance on complex infrastructure, including numerous cameras and sensors installed throughout the store.

In contrast, many developing regions face financial and infrastructural limitations that make such solutions impractical. Therefore, there is a strong need for alternative systems that can deliver similar benefits while remaining affordable and scalable.

The SCAN_GO system is proposed as a practical solution to bridge this gap. By utilizing an embedded smart shopping basket equipped with an Android tablet, an ESP32 microcontroller, a barcode scanner, and a weight sensor, the system enables customers to seamlessly scan products and complete secure digital payments. This localized hardware approach eliminates the need for expensive store-wide infrastructure while maintaining the core advantages of cashier-less shopping.

1.2 Problem Statement

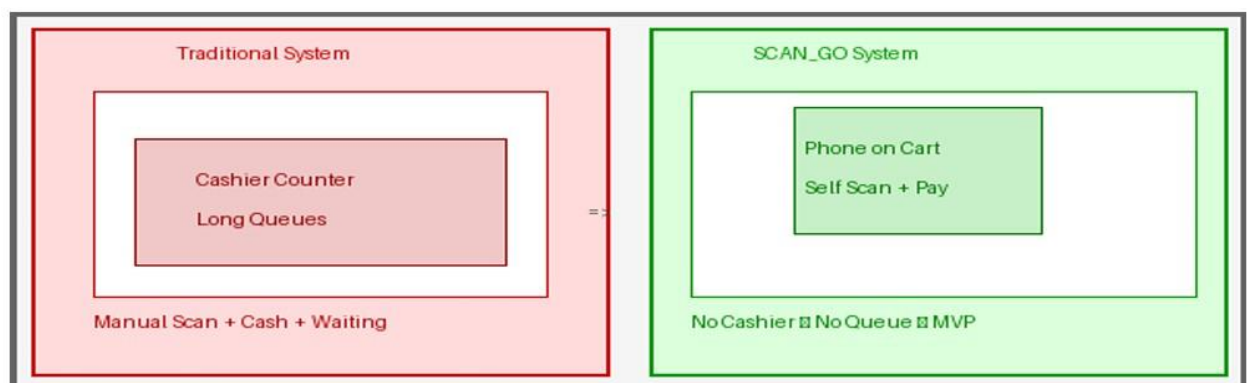


Figure 1.1 Comparison between traditional checkout and SCAN_GO system

Despite the rapid advancement in technology and the increasing adoption of digital solutions across various industries, traditional supermarket systems still rely heavily

on outdated checkout mechanisms. These systems, although functional, present several limitations that negatively impact both operational efficiency and customer satisfaction.

Supermarkets are designed to handle a large number of customers daily, especially during peak hours such as weekends and holidays. However, the traditional checkout process becomes a bottleneck in the overall shopping experience. Customers are required to wait in long queues, interact with cashiers, and go through repetitive manual procedures before completing their purchases. This process is not only time-consuming but also inefficient in handling high customer volumes.

Moreover, the reliance on human intervention introduces variability and inconsistency in service quality. The performance of the checkout process depends largely on the speed and accuracy of cashiers, which can vary from one employee to another. This inconsistency can lead to delays, errors in billing, and overall dissatisfaction among customers.

In addition, traditional systems are not well-equipped to meet the expectations of modern consumers who demand speed, convenience, and seamless experiences. With the widespread use of smartphones and digital payment solutions, customers increasingly prefer self-service options that allow them to complete transactions quickly and independently.

Another critical issue is the lack of real-time data accessibility. In conventional systems, store managers often rely on delayed reports to monitor sales and inventory levels. This delay in information flow affects decision-making processes, leading to inefficient stock management, overstocking, or stock shortages.

Furthermore, the dependency on cash transactions introduces multiple operational challenges. Handling physical cash requires additional security measures, increases

the risk of human error, and slows down the transaction process. It also creates inconvenience for customers who prefer digital payment methods.

Security is also a major concern in traditional retail environments. Without an automated, cross-referenced verification system, supermarkets are highly vulnerable to inventory shrinkage caused by theft or unpaid items. This results in significant financial losses and reduced profitability.

To better understand the challenges faced by traditional supermarket systems, the problem can be categorized into several key areas:

1.2.1 Long Waiting Times at Checkout

One of the most visible and frustrating problems in traditional supermarkets is the extensive waiting time at checkout counters. During peak hours, customers often spend significantly more time waiting in lines than actually selecting their items.

This bottleneck arises from several systemic limitations, including a restricted number of operational checkout lanes, slow manual processing of individual items, and high customer volumes. Long queues drastically reduce customer satisfaction and degrade the overall retail experience.

In many instances, excessive delays lead to cart abandonment, resulting in direct sales losses for the store. Furthermore, checkout congestion creates physical crowding near the store exits, disrupting the overall traffic flow within the supermarket. By shifting the checkout process directly to the smart basket, this localized approach completely decentralizes the transaction point and eliminates the need for physical queues.

1.2.2 Inefficiency of Manual Checkout Processes

The traditional checkout workflow relies entirely on human intervention to manually scan barcodes, verify prices, process payments, and print physical receipts. This repetitive and labor-intensive routine inherently acts as an operational bottleneck.

Heavy reliance on manual handling escalates the probability of human errors, including pricing discrepancies, billing mistakes, or missed items. Furthermore, these sequential manual procedures cap the maximum throughput of the store, limiting its capacity to handle large customer volumes efficiently and driving up labor costs. Shifting this workflow to an automated, decentralized system via the smart basket allows for parallelized transaction processing, drastically boosting overall retail efficiency.

1.2.3 Complications Associated with Cash Transactions

Physical cash transactions introduce considerable friction and operational delays into the retail checkout workflow. Cashiers must manually count currency, calculate and dispense change, and verify banknote authenticity, which prolongs processing time per customer.

Furthermore, cash handling escalates the risk of human accounting errors, bookkeeping discrepancies, and security vulnerabilities such as internal theft or counterfeit fraud. Managing physical tender also imposes heavy administrative overhead on store management, including daily reconciliation, secure physical storage, and armored transport deposits.

From the consumer perspective, cash payments lack flexibility and convenience, particularly when managing change. By integrating direct digital payment gateways into the smart basket's internal software, the system eliminates cash-related overhead, streamlines financial auditing, and secures the transaction lifecycle.

1.2.4 High Operational Costs

Sustaining a conventional storefront architecture demands substantial financial expenditure to recruit, train, and manage a large cohort of cashiers. These fixed and variable labor costs—encompassing salaries, onboarding resources, and supervisory overhead—substantially deflate net profit margins.

Furthermore, under the traditional retail model, scaling operations to meet fluctuating consumer demands exhibits a linear dependency; spike periods require opening additional physical lanes, thereby compounding operational expenses. This resource-intensive framework limits long-term fiscal sustainability, especially for enterprises striving to achieve lean operations and cost optimization. By decentralizing the point of sale via the smart basket, the store can seamlessly handle high throughput without a proportional increase in personnel overhead.

1.2.5 Lack of Real-Time Inventory and Sales Monitoring

Conventional retail architectures fail to deliver instantaneous visibility into live store operations, forcing management to rely on delayed, periodic batch reports to evaluate sales velocity and inventory levels. This chronic latency in data synchronization severely impedes proactive decision-making, culminating in critical operational friction such as localized stock shortages, lost sales opportunities, or excessive overstocking.

Without access to highly granular, real-time transaction streams, maintaining an optimal inventory equilibrium becomes logistically unfeasible. By leveraging a network of smart baskets that instantly push telemetry data—such as immediate barcode validation and item subtractions—to a centralized cloud backend, the system bridges this data gap. This enables automated, real-time inventory reconciliation and predictive stock replenishment alerts.

1.2.6 Inventory Shrinkage and Security Issues

Inventory shrinkage represents a critical financial vulnerability within traditional retail environments, driven by systemic theft, product damage, and administrative discrepancies. Conventional storefront frameworks lack an automated, cross-referenced mechanism to verify that every item in a customer's physical possession has been accurately scanned and processed prior to exit.

This absence of real-time point-of-sale control significantly escalates retail loss and dampens net profitability. Furthermore, relying on manual surveillance to monitor high customer volumes is structurally inefficient and highly prone to human oversight. By embedding an automated weight-verification engine—driven by the \$HX711\$ load cell module—directly into the smart basket, the system continuously audits cart weight against scanned barcode metadata. This localized hardware validation, paired with a secure QR-code-controlled exit gate, constructs an objective, tamper-resistant security layer that mitigates theft without operational friction.

1.2.7 Poor Customer Experience

All the previously mentioned bottlenecks collectively degrade the overall customer experience. Chronic checkout delays, outdated manual scanning, and rigid payment workflows foster consumer frustration and dissatisfaction. In today's highly competitive retail market, customer experience is a primary determinant of business retention and brand loyalty. Supermarkets that fail to meet modern convenience expectations risk losing substantial market share to competitors leveraging automated infrastructure. By decentralizing the checkout process through an interactive, basket-integrated interface, the system transforms a high-friction chore into a seamless, autonomous customer journey.

1.2.8 Limitations of Existing Smart Retail Solutions

Although pioneering computer-vision-based solutions like Amazon Go have validated the feasibility of cashier-less environments, their ubiquitous deployment remains constrained by prohibitive capital expenditure and intricate infrastructural prerequisites. These high-end ecosystems rely heavily on dense ceiling-mounted camera grids, distributed weight-sensing shelves, and continuous sensor fusion algorithms.

Consequently, the localized computing overhead and high maintenance costs make them economically unviable and technically impractical for small-to-medium supermarkets, particularly within developing economies. This creates an immediate market gap for a localized, edge-computed alternative. There is a critical need for an affordable, infrastructure-light solution that architecture-wise shifts the intelligence from the building itself to the individual shopping vessel, achieving frictionless checkout without structural store modifications.

1.2.9 Summary of the Problem

In summary, conventional retail ecosystems are constrained by a network of interconnected operational challenges, predominantly characterized by throughput inefficiencies, escalating labor overhead, compromised customer satisfaction, and a critical deficit in real-time data synchronization. These systemic friction points underscore the urgent necessity for an innovative, infrastructure-light solution capable of modernizing store operations while maintaining fiscal and scalable viability. The proposed SCAN_GO system directly addresses these multi-dimensional vulnerabilities, introducing a decentralized, sensor-integrated shopping basket that serves as a cost-effective, high-efficiency alternative to traditional checkout frameworks.

1.3 Aim of the Project

The primary objective of this project is to design, develop, and evaluate the SCAN_GO system—an infrastructure-light, decentralized smart supermarket checkout platform engineered to eliminate traditional checkout bottlenecks and replace them with an autonomous, user-centric, and hardware-integrated framework. The project aims to transition conventional retail operations into a digitized self-service experience by shifting the computing intelligence directly onto the individual shopping basket. By combining an Android tablet interface, an ESP32 microcontroller, a localized barcode scanner, and an \$HX711\$-driven load cell weight verification module, the system aims to automate item auditing and digital payment processing, thereby minimizing reliance on human intervention while preserving strict inventory security.

Furthermore, this project is fundamentally designed with cost-effectiveness and scalable deployment in mind. Unlike capital-intensive computer-vision frameworks, SCAN_GO focuses on utilizing accessible edge devices to deliver high-tier automation, making it structurally and financially viable for supermarkets in developing regions with infrastructure constraints. On an operational level, the system aims to optimize store resource allocation; automating data ingestion via edge hardware reduces administrative staff workload, allows human capital to redirect toward value-added tasks, and drives down overhead costs. For the consumer, the system targets a completely frictionless journey by enabling concurrent item scanning, real-time telemetry of total expenditure, and instantaneous digital transaction processing. Finally, the project aims to empower store management through real-time telemetry of sales and inventory dynamics to enable predictive replenishment, while deploying a secure, QR-code-activated exit gate to mitigate inventory shrinkage and validate transactional integrity.

1.4 Project Objectives



Figure 1.2 SCAN_GO – Smartphone mounted on shopping cart

To realize the overarching aim of the SCAN_GO platform, a set of strategic, measurable, and engineering-centric objectives has been defined. These objectives serve as the technical milestones guiding the architecture design, hardware-software integration, and empirical evaluation of the system.

1.4.1 Development of a Mobile-Based Self-Scanning System

A foundational objective of this project is to develop a high-performance Android application deployed directly onto a tablet interface mounted on the smart shopping cart. The user interface is engineered to provide an intuitive, seamless human-machine interaction, allowing customers to autonomously manage their shopping lifecycle.

The scanning module interfaces natively with localized hardware scanners to achieve rapid, low-latency barcode ingestion. This sub-system guarantees high-precision item identification, immediately appending verified products to the user's localized digital cart. To enhance transparency and shopping flow, the application architecture executes real-time data binding to render dynamic telemetry, including precise unit pricing, localized item quantities, cumulative cart subtotals, and instantaneous promotional data updates.

1.4.2 Implementation of a Digital Payment System

Another core objective involves embedding a secure, low-latency digital payment architecture directly within the application layer. The system is designed to facilitate instantaneous point-of-sale transactions through various electronic payment methodologies, including tokenized credit/debit cards, decentralized mobile wallets, and secure online payment gateways.

The transactional pipeline prioritizes computational efficiency and cryptographic security to minimize checkout latency. To preserve data integrity and comply with financial security standards, the subsystem implements advanced security protocols, including End-to-End Encryption Secure Sockets Layer (SSL/TLS) communication channels, and tokenization. These operational safeguards protect sensitive user credentials, prevent unauthorized network access, and mitigate transaction-level fraud.

1.4.3 Design and Development of a Backend System

A critical objective of this project is to architect and deploy a robust, scalable centralized backend system designed to govern the platform's core business logic, real-time data processing, and relational storage. The backend engineering focuses on constructing a unified ecosystem that orchestrates seamless, bidirectional communication between the basket-integrated Android interface, the ESP32 edge hardware, and the centralized administrator dashboard via structured Application Programming Interfaces.

The backend infrastructure is engineered to optimize throughput and minimize latency when managing highly concurrent data streams, including cryptographic user credentials, extensive product metadata databases, and immutable financial

transaction records. To achieve long-term scalability and operational reliability, the system architecture decouples data ingestion from relational query processing, utilizing optimal database indexing and secure communication protocols to prevent race conditions and ensure strict data synchronization across all system nodes.

1.4.4 Creation of an Administrative Dashboard

A critical objective of this project is to design and implement a web-based administrative dashboard that serves as the centralized command-and-control center for retail management. The dashboard's architecture focuses on consolidating high-velocity data streams pushed from the backend, providing store operators with real-time visibility into distributed checkout analytics, inventory fluctuations, and concurrent customer transaction cycles.

The front-end interface is engineered around interactive data visualization frameworks, incorporating responsive analytical charts, multi-dimensional reporting tools, and low-latency anomaly alerts. This design empowers management with a data-driven decision-making environment to monitor localized store performance and track hardware fleet activity. To ensure enterprise-grade security and operational integrity, the platform integrates Role-Based Access Control protocols, restricting system visibility and configuration rights strictly to authenticated and authorized personnel.

1.4.5 Implementation of QR Code-Based Verification

To preserve transactional integrity and enforce strict egress security, the system architecture incorporates a dynamic cryptographic QR code generation engine within the application layer. Upon the execution of a successful financial transaction, the backend database generates an immutable digital receipt encrypted into a localized QR code token.

This token serves as a tamper-resistant proof of purchase containing structured cart metadata. At the storefront perimeter, a dedicated hardware-integrated reader scans the generated QR code to cross-reference customer transaction data with the cloud registry. This automated verification mechanism triggers the physical gate actuation, eliminating unauthorized product removal and effectively mitigating inventory shrinkage. By replacing subjective manual exit checks with objective hardware access control, this subsystem constructs a reliable, high-throughput security perimeter that protects both retail assets and consumers.

1.4.6 Integration of System Components

A compounding objective of this project is to orchestrate seamless end-to-end integration and interoperability across all distributed system components. The architecture is engineered to transform the standalone layers—the basket-integrated Android interface, the ESP32 embedded edge hardware, the centralized backend framework, the cloud database, and the web-based administrative dashboard—into a unified, cohesive retail automation ecosystem.

This objective necessitates the rigorous design and implementation of lightweight Application Programming Interfaces (APIs) and optimized communication protocols. By establishing secure, bidirectional JSON-payload data pathways the system architecture ensures low-latency data synchronization, eliminates race conditions during concurrent cart updates, and guarantees fluid telemetry flow from the physical shopping vessel to the administrative dashboard.

1.4.7 Ensuring System Scalability

The system should be designed in a way that allows it to scale efficiently as the number of users and transactions increases. This includes the ability to handle multiple stores, large product databases, and high traffic volumes.

Scalability ensures that the system can grow with the business and adapt to future requirements.

1.4.8 Secure and Efficient Inventory Management

The system should provide accurate and real-time data on inventory levels, enabling better stock management. This helps prevent issues such as overstocking or stock shortages.

Efficient inventory management contributes to improved profitability and customer satisfaction.

Security is a critical aspect of the system. Measures must be implemented to protect user data, secure transactions, and prevent unauthorized access.

The system should also be reliable, ensuring consistent performance without failures or downtime.

1.5 Proposed System Overview

The SCAN_GO platform introduces an innovative, decentralizing architecture engineered to overcome the throughput bottlenecks, high operational overhead, and infrastructure constraints inherent in traditional retail checkout models. Distinct from high-cost, computer-vision frameworks that require pervasive environmental modifications, SCAN_GO shifts the system's computational intelligence directly onto the individual shopping vessel. By utilizing an edge-computing tablet mounted on the shopping cart interfaced with localized sensors, the platform achieves autonomous checkout capability with high fiscal and scalable viability.

- **Initialization and Edge Data Ingestion:** The shopping journey begins at a cart integrated with an Android-based terminal coupled to an ESP32 microcontroller network. As items are selected, the customer utilizes the localized barcode scanning

module to execute rapid serialization. The application layer performs asynchronous API calls to the centralized backend database to instantly retrieve and bind product metadata, including item nomenclature, unitary cost, and localized stock availability.

- **Real -Time Data Binding & Audit Trails:** Concurrently, each validated product updates a local digital cart instance. The front-end user interface renders real-time telemetry of cumulative expenditure, itemized sub-totals, and specific quantities. This continuous data visibility eliminates transactional ambiguity, empowering consumers with precise financial control before approaching the storefront perimeter.
- **Decentralized Checkout Execution:** A core advantage of this architecture is the distribution of transaction processing throughout the shopping temporal sequence, eliminating the traditional congestion points at centralized checkout lines. Upon cart completion, the payment phase is executed natively within the application layer via an authenticated, encrypted digital payment pipeline. Successful transaction resolution triggers the backend server to generate an immutable, tamper-resistant cryptographic QR code token representing the digital receipt metadata.
- **Access Control Verification & Analytics:** At the storefront egress point, a hardware-integrated optical scanner evaluates the cryptographic QR token against the cloud registry. Validated receipts automatically actuate the physical security gate, mitigating inventory shrinkage while preserving high-throughput exit dynamics. Simultaneously, all transaction telemetry is ingested into the web-based administrative dashboard, equipping store management with real-time operational analytics for predictive replenishment and data-driven inventory control.

1.6 System Architecture Overview

The architecture of the SCAN_GO platform is engineered to ensure high throughput, horizontal scalability, and seamless interoperability across heterogeneous nodes. The system implements a decoupled, centralized-broker architecture model, wherein a high-performance backend server acts as the central orchestration layer to manage asynchronous data processing, state synchronization, and peripheral coordination.

The structural design is organized into a multi-tiered, layered architecture. This decoupling enforces strict modularity and separation of concerns (\$SoC\$), ensuring that individual system components—ranging from embedded hardware abstraction layers to cloud-hosted databases—can be developed, debugged, and scaled independently. This modular framework minimizes systemic dependencies, reducing maintenance overhead and providing a highly adaptable foundation for future enterprise retail integrations.

1.6.1 Mobile Application Layer

The client-side application layer acts as the primary human-machine interface for the customer, serving as the central orchestration point for user-system interaction. Deployed directly on the basket-integrated Android terminal, this layer is structurally responsible for executing localized edge interactions, including synchronous barcode data capture, stateful cart memory management, and secure cryptographic payment processing.

The application architecture utilizes reactive data-binding frameworks to deliver deterministic, low-latency visual feedback, ensuring immediate interface state updates following any peripheral hardware trigger.

A foundational sub-module within this layer is the barcode ingestion engine. Interfacing natively with the cart's hardware scanner peripheral via localized

communication APIs, the subsystem translates raw optical captures into structured numerical payloads for backend validation. This pipeline is optimized for high-precision, sub-second decoding to eliminate transactional friction.

Structurally, the application layer encapsulates state management routines that enable users to execute the following core operations:

Dynamic Inventory Auditing: Instantiate and render asynchronous views of the local digital cart contents.

Stateful Quantity Mutation: Modify item quantities natively with automatic local verification.

Real-Time Expenditure Telemetry: Stream continuous, real-time computations of the cumulative cart cost, including discounts and tax parameters.

Transactional Pipeline Egress: Formally compile cart metadata and securely handover session control to the integrated digital payment gateway.

1.6.2 Backend System Layer

The backend system is the core component that manages all operations within the SCAN_GO system. It handles business logic, processes requests, and ensures communication between different components.

The backend is responsible for:

- Managing product data
- Processing transactions
- Handling payment verification
- Generating QR codes
- Storing user activity

It acts as a bridge between the mobile application and the database, ensuring that all data is processed efficiently and securely.

The backend must be designed to handle multiple concurrent users without performance degradation. This requires efficient data handling and optimized APIs.

1.6.3 Database Layer

The database layer is responsible for storing and managing all system data. This includes product information, transaction records, user data, and inventory details.

The database must be structured in a way that allows fast data retrieval and efficient updates. It should also ensure data consistency and integrity.

Key data stored in the database includes:

- Product catalog
- Pricing information
- Transaction history
- Inventory levels

1.6.4 Admin Dashboard Layer

The admin dashboard provides store managers with a comprehensive view of store operations. It allows them to monitor sales, track inventory, and analyze customer behavior in real time.

The dashboard includes various features such as:

- Sales reports

- Inventory tracking
- Transaction monitoring
- Data visualization

This layer plays a crucial role in decision-making, enabling managers to respond quickly to changes in demand and operational conditions.

1.6.5 Communication Layer (API Integration)

Communication between system components is handled through APIs (Application Programming Interfaces). The backend exposes endpoints that allow the mobile application and admin dashboard to interact with the system.

The communication is typically carried out over secure protocols such as HTTPS to ensure data protection. APIs must be well-designed to ensure reliability and performance.

1.6.6 Security Considerations in Architecture

Security is a fundamental aspect of the system architecture. Measures must be implemented to protect user data and ensure safe transactions.

These measures include:

- Data encryption
- Secure authentication
- Access control mechanisms
- Protection against unauthorized access

Additionally, the QR code verification system adds an extra layer of security by ensuring that only paid items leave the store.

1.6.7 Scalability and Performance

The system architecture is designed to support scalability. As the number of users increases, the system should be able to handle higher loads without affecting performance.

This can be achieved through:

- Efficient backend design
- Optimized database queries
- Scalable cloud infrastructure

Scalability ensures that the system remains effective as the business grows.

1.6.8 Summary of System Architecture

In summary, the SCAN_GO system architecture is designed to provide a robust, scalable, and efficient solution for smart retail operations.

The integration of mobile technology, backend processing, and real-time data management creates a unified system that enhances both customer experience and operational efficiency.